

EXACT MEMORY-TIME OPTIMIZATION FOR PREFIX-CACHED LANGUAGE MODEL SERVING

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Prefix-Certificate Retention (PCR) • September 2026

A usable cache hit is a set of prerequisites.

Represent those prerequisites explicitly: exact static timeout optimization becomes one minimum cut.

1. Model and construction

A request can reuse a block only when its entire preceding prefix is available. Blocks have grouped, reset-on-access timeouts; storage is charged over a finite observation window.

Maximize usable prefix blocks

minus $\lambda \times$ occupied block-seconds

Threshold nodes: choosing a longer timeout incurs incremental storage cost.

Hit nodes: a reward requires its local timeout threshold and the previous prefix-hit node.

Maximum closure: every selected node includes its prerequisites. Standard minimum cut gives the optimum.

Other requests can refresh ancestors, so longer descendant timeouts can remain useful.

2. Mathematical and empirical results

Exact finite-grid optimum

One minimum cut; graph size is linear in block lookups and timeout choices.

No discretization error required

Every timeout can be rounded down to a training reuse age without losing a hit or increasing cost.

Certify the simpler policy

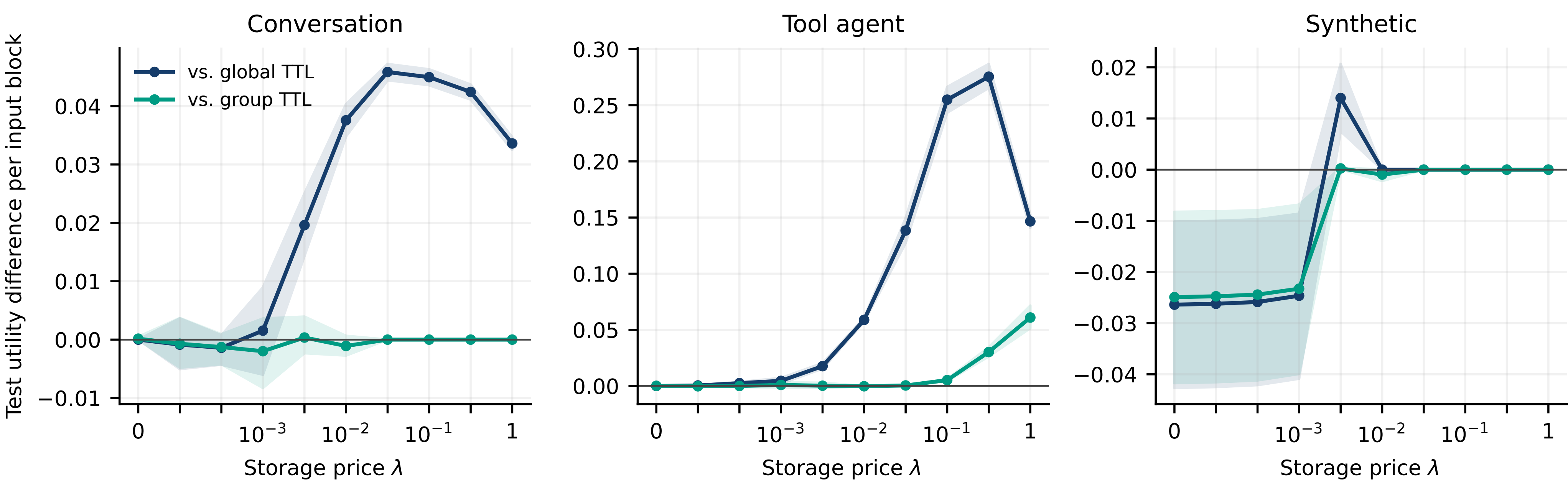
Coherent optimum $\leq$ exact optimum $\leq$ independent local-hit upper bound.

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Native fixed-grid cases where ordered timeouts already attain the unrestricted training optimum.

3. Chronological replay of 39,632 public requests

Mooncake FAST'25: conversation, tool-agent and synthetic workloads. Training / validation / test: 40 / 20 / 40% of elapsed time. All ten declared storage prices are reported.



Test utility: usable prefix blocks minus storage price times occupied block-seconds, normalized per input block. Lines compare the validation-selected coherent policy with global and group timeouts. Bands are descriptive 60-second-bucket bootstrap intervals; they are not deployment-wide guarantees.

Use: an offline configuration oracle and retention-policy audit.

11 tests include exhaustive profiles, independent event replay and 450 full-policy LP comparisons. This is cache replay with immediate post-batch availability, not measured GPU latency or a hard-capacity guarantee.

Sources: Picard (1976); Mooncake (FAST25); Kareto (2026).

Code, proofs, data provenance and complete results:

github.com/shi1720/prefix-certificate-retention

